

- for data transfer by using a specific TCP port 1888
 - No e-mail access
 - No access to the customer vault from other PCs in the office
- Apart from these customer requirements, Shree also finalised a firewall policy to secure and control Internet access. The policy mentioned the following additional requirements for the office network:

- Website filtering
- Blocking access to free Web mail sites (viz., Yahoo!, Hotmail, Gmail, etc)
- E-mails to be sent and received only using the company mail server
- VPN access for two project leaders from their laptops
- Time synchronisation on all PCs

According to the requirements and in consultation with Kedar, a network diagram for the firewall implementation was prepared (Figure 1).

To Shree's surprise, the basic installation of IPCop was very simple and straightforward.

Implementation steps

The first and foremost step was to find an old PC for the IPCop and add two 10/100 Ethernet cards (one was on-board). The next steps were to download the IPCop ISO file, burn the image and install it with RGB configuration—red stood for the Internet, green for the protected internal subnet #1 for office PCs, and blue for the protected internal subnet #2 for the customer's PCs.

After following the plain straightforward installation steps, IPCop was working in less than one hour! The PCs being used for the customer's work were immediately separated from the other office PCs. The only change required to start time synchronisation was to set the IPCop IP address in the system clock of all the PCs in the organisation.

The default installation of IPCop does not support restricting Internet access to the customer vault, blocking websites, blocking outgoing traffic and OpenVPN. Since these were a must to satisfy the customer's requirements as well as the security policy, some of the IPCop add-ons were required. Here, Shree's friend Kedar was of great help. Three add-ons were suggested and configured:

- Advanced proxy + URL Filter** for filtering unwanted websites. URL Filter displays a set of block categories. By ticking the checkbox of the category, websites in the corresponding block are denied access. The list of sites in these categories can be scheduled to be updated automatically every day, week or month, from various download sources.
- Zerina** for OpenVPN support. Shree's company had an Internet connection based on the PPPoE dynamic IP address. Therefore, VPN support appeared to be impossible, since a static IP is required for most of the VPN servers. Here, DDNS service supported by IPCop (by default) came in handy. IPCop has an in-built certificate generator required to create VPN connections. Certificates

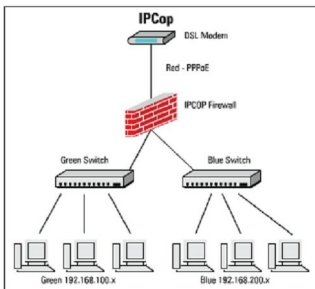


Figure 1: The network diagram for the firewall implementation

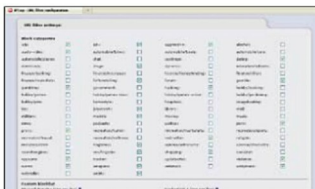


Figure 2: Filtering block categories

were generated and distributed to the corresponding clients. The VPN configuration was completed on client notebooks by installing OpenVPN GUI.

- Block Outgoing Traffic (BOT)** was installed for controlling port-wise Internet access. The BOT configuration was really tricky and required several trials before the final settings were completed. The configuration was done in four steps:
 - Grouping internal PCs as per the required services access
 - Grouping the external IP addresses required to be accessed from the predefined internal groups
 - Defining and grouping the various services required to be accessed
 - Creating the firewall rules so that the internal groups can access the external groups by the required services groups

To elaborate on these three steps, let's consider an actual example of restricting customer vault access:

- PCs in the blue network were grouped as *customerpcs*.
- customer vault* was defined by its IP address.
- The service named *customer vault access* was defined